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Paleomagnetic and Rock Magnetic Study of Combustion Metamorphic Rocks in Israel

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Paleomagnetic and rock magnetic studies were combined with the mineralogy and metamorphic petrology of a sequence of combustion metamorphic (CM) rocks in Israel. Samples from 12 sites from the three major outcrops of the Hatrurim Formation ("Mottled Zone") were analyzed. These are originally Cretaceous bituminous chalks and marls which have undergone spontaneous combustion during the Late Neogene. The analyses included natural remanent magnetization (NRM) intensity, alternating field (AF) and thermal demagnetization, and isothermal remanent magnetization (IRM) measurements. Mineralogical phases were identified by X-ray diffraction and electron microprobe. All metamorphosed rocks have a NRM intensity far above normal sedimentary rocks. This increase in intensity is observed even in rocks where no mineralogical indications of metamorphism are observed. The major contributors to the rock magnetism in these rocks are maghemite, hematite and rare magnesioferrite. The magnetic properties were acquired during the burning process as a consequence of oxidation of Fe-sulfides. The in situ direction of the mean magnetic vector is Dec = 8.2°, Inc = 49.2°, K = 40 and α_{95} = 5.2 whereas the stratigraphically corrected mean is Dec = 25.1°, Inc = 46.9°, K = 26 and α_{95} = 6.5. The stratigraphic mean direction is in accordance with the Pliocene expected field direction for this area as derived from stable Africa poles. Therefore combustion must postdate tectonic tilt or vertical axis rotation associated with Late Cretaceous major folding and faulting and rifting associated with the Dead Sea Transform. Normal and reverse polarity and paleosecular variation indicate that combustion at the different sites was not simultaneous.

INTRODUCTION

High-temperature, low-pressure (sanidinite facies) metamorphism of sedimentary rocks has been documented in several cases of surface spontaneous combustion of organic matter rich protoliths, such as coal, oil sands, or oil shales [*Reverdatto*, 1970, *Mathews and Bustin*, 1984, *Bentor et al.*, 1981, *Jones et al.*, 1984]. The process has been termed combustion metamorphism (CM). This thermal metamorphic material can be strongly magnetic and thus the source for surface magnetic anomalies [*Hasbrouck and Hadsell*, 1976; *Watson*, 1978; *Jones et al.*, 1984; *Cisowski and Fuller*, 1987; *Cisowski*, 1988]. When natural combustion occurs in bituminous marine sediments, the temperature increase and the ignition of the organic material is often the consequence of low temperature oxidation of finely disseminated framboidal pyrite, a frequent component of such rocks [*Mathews and Bustin*, 1984]. Since oxygen is a vital component in burning, CM generally affects rocks at or near the surface. CM is usually associated with open fractures caused by brittle deformation such as regional faulting and folding.

Postdepositional, low temperature oxidation of pyrite has been lately recognized as an important remagnetization process in limestone [*Suk et al.*, 1990; *Reynolds*, 1990]. As in the case of low-temperature oxidation, the metamorphic-magnetic mineralogy of the CM rocks is the product of the primary iron-bearing minerals, the availability of oxygen, the temperature, and the presence of other minor metallic impurities.

The "Mottled Zone" (MZ) complex (formally termed the Hatrurim Formation) is a sequence of CM rocks in Israel and Jordan, cropping out in several localities, the largest of which are the Hatrurim basin (near Arad), the Maale Adumim area on the road from Jerusalem to Jericho, and the Daba-Siwaqa area south of Amman (Figure 1). The unmetamorphosed protoliths were the bituminous rocks of the Cretaceous Ghareb and Early Tertiary Taqiye Formations. These are well bedded bituminous chalks and marls, and some phosphorites. Rocks of both of these formations contain large amounts of iron sulfides, both as dispersed microcrystalline pyrite and as pyrite concretions, up to several centimeters in diameter. The typical natural remanent magnetization (NRM) intensity of such protoliths is on the order of 10^{-4}A/m; the detrital component in these rocks is negligible, and no significant amount of detrital magnetite has been detected in them.

The metamorphic rocks are characterized by the disappearance of sedimentary bedding, strong veining, and "mottling" in color; red-green-yellow-black staining is typical.

The initial, intuitive interpretation of the origin of these rocks as due to CM [*Lees*, 1928; *Bentor and Vroman*, 1960], has since been supported by extensive mineralogical and petrological studies [*Bentor et al.*, 1963; *Gross*, 1977; *Kolodny and Gross*, 1974; *Matthews and Gross*, 1980].

A mineral assemblage which includes numerous high-T, low-P phases has been identified. The abundant rock forming minerals in this assemblage are, besides (the relict protolithic) calcite, the metamorphic minerals such as spurrite, larnite, gehlenite,

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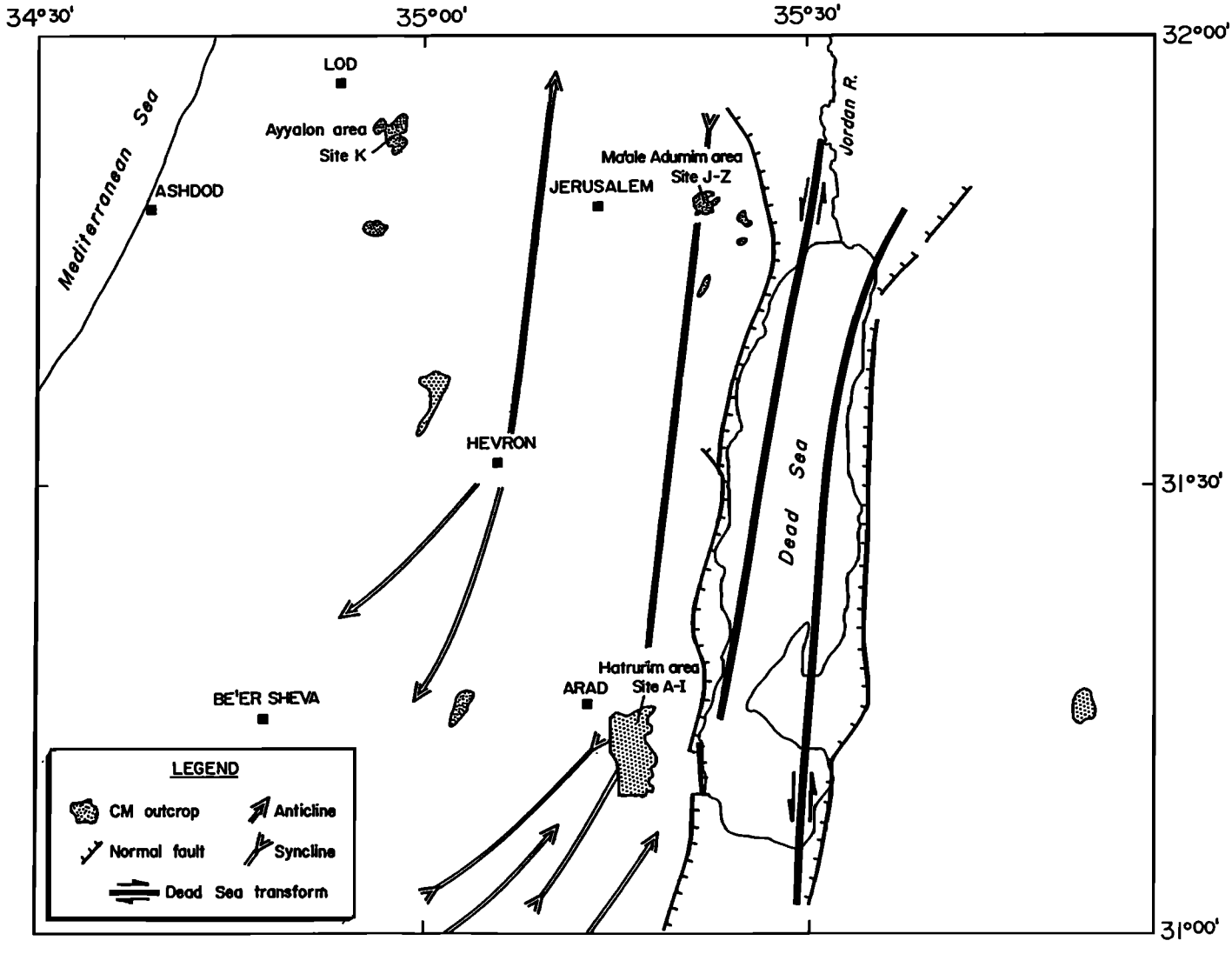


Fig. 1. Location map showing major structural elements, the main CM outcrops, and sampling sites.

diopside, wollastonite and numerous others. Elsewhere, such assemblages are usually typical of narrow contact zones between igneous intrusions and siliceous carbonate country rocks. In our case, there is ample proof that (a) no intrusion has been observed, (b) metamorphism is confined to the surface where oxygen supply for burning was abundant, and (c) that massive burning of bituminous matter has been recorded by a strong enrichment in ^{12}C and ^{16}O [*Kolodny and Gross*, 1974]. The loss of bedding, change in color and abundant veining, mark metamorphism in the field. Calcite remains the major phase in the slightly metamorphic rocks. This relict mineral can be distinguished from the nonmetamorphic equivalents merely by different degrees of recrystallization and quite often by staining. The changes in color are probably due to changes in the mineralogy of the minor, transition metal bearing phases. In unmetamorphosed rocks the bulk of the iron (≈1-4 wt % in whole rock) occurs as sulfides. The bituminous rocks are also enriched in Ni and Cr. The gradual oxidation of Fe, which has not been mobilized, as well as the oxidation and subsequent mobilization into veins of Cr and Ni, are the most visually striking aspects of metamorphism.

The typical medium-grade rock forming mineral in the MZ is spurrite, with minor brownmillerite. In these rocks the Fe-phase is in most cases an oxide, usually maghemite. Relict greigite has been observed in some black colored spurrites. With increasing grade, larnite becomes a significant phase.

The highest grade common rocks in the MZ are gehlenite-garnet (shorlomite)-pyroxene fels rocks. These contain hematite, often magnetite, and, in one case, Ni or Ti bearing chromite.

Fission-track dating of the Maale Adumim outcrop [*Kolodny et al.*, 1971] suggests that burning at this site took place in the Middle Miocene (13.6 ± 2Ma). Stratigraphic evidence suggests that metamorphism at Hatrurim basin is younger than the Miocene Hazeva Formation [*Kolodny et al.*, 1973]. Detailed mapping, preliminary K-Ar dating (D. Zur, personal communication, 1991) and preliminary electron spin resonance dating (N. Porat, personal communication, 1991) suggest that the burning in the Hatrurim area has been even younger, maybe late Pliocene [*Burg et al.*, 1992]. A more detailed understanding of the metamorphic decarbonation process has been gained by isotopic analysis of carbon and oxygen in MZ carbonates and silicates [*Kolodny and Gross*, 1974; *Matthews and Kolodny*, 1978] and the mineralogy of apatite [*Matthews and Nathan*, 1977]. The results of these studies suggest that for large parts of the metamorphosed complex the prevailing temperatures were in excess of 300°-400°C. In the

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hotter areas, temperatures higher than 800°C have probably been reached.

The two large CM fields in Israel, the Hatrurim basin and Maale Adumim, are both characterized by large positive aeromagnetic anomalies [*Folkman and Yuval*, 1976]. An analysis of the rock magnetic properties of the CM sequence may provide further constrains on the process of combustion metamorphism. The distribution of temperatures in the burning rock pile, the time during which the magnetic properties were acquired, the synchroneity of the burning process in different outcrops, as well as the time relation between folding and metamorphism, are all problems upon which paleomagnetic and rock magnetic measurements may shed some light.

SAMPLING AND METHODOLOGY

Most of the samples came from the three major outcrops of the MZ: the Hatrurim area (near Arad, sites A-I), the Ma'ale Adumim area (northern Judea desert, sites J, Z), and one site from the Valley of Ayyalon outcrop (site K) (Figure 1). All samples were collected as oriented, 2.54-cm-diameter, field drilled cores using both magnetic and sun compasses for orientation. Standard specimens, 2.54 cm long were prepared from the cores. The same cores were used for electron microprobe (EPM) and X ray diffraction (XRD) analyses. In the Hatrurim valley one location (sites A, B, C, D, E, and H) provided the entire spectrum from nonmetamorphic rocks, through low- to medium- to high-grade CM rocks (Table 1). The criterion for establishing the degree of metamorphism was here a purely mineralogical and textural one. At another location, sites G and I provided what appears as nonmetamorphic phosphorite and limestone concretions and at site F low CM rocks were sampled (Table 1). Sampling sites were chosen to avoid slump and collapse structures, and the bedding attitude was carefully measured wherever possible.

Paleomagnetic measurements were performed at Stanford University and at the Institute of Petroleum Research and Geophysics in Holon, Israel. Remanence measurements and demagnetization procedures were conducted using automated cryogenic magnetometers with integrated alternating field (AF) demagnetization coils and commercially available ovens for thermal demagnetization. The thermal demagnetization experiment failed on several gehlenite-bearing samples from site H. Upon heating, the specimen exploded and disintegrated probably due to the presence of zeolites. Isothermal remnant magnetization (IRM) acquisition experiments were performed on selected samples. Directions of all components of magnetization were determined using orthogonal vector plots and *Kirschvink's* [1980] method. Statistical data analysis has been carried out following *Fisher* [1953].

All rocks were examined in thin sections, the major phases identified by XRD, and the chemical composition of the different phases was determined by EPM (JEOL JXA-8600 Superprobe).

RESULTS AND INTERPRETATION

Rock Magnetism and Paleomagnetism

The CM rocks show a large range of NRM intensity over about 4 orders of magnitude ($2*10^{-4}$ up to 6 A/m) (Table 1). The highest NRM intensity values are far above normal NRM intensity of sedimentary rocks and are comparable to the NRM's of continental basalts. No correlation can be made between the CM grades or rock color and the NRM intensity (Table 1). Samples from sites B, C, G, and I, which by field and mineralogical criteria are nonmetamorphic, also show this large range of NRM intensity (Table 1). Thus, applying magnetic criteria (NRM intensity, analysis of magnetization direction, (Table 2), and demagnetization behavior (as discussed below), it appears that all the rocks sampled, except those from site A, did undergo some degree of CM. The above mentioned lack of correlation should be considered in view of the finding of *Cisowski and Fuller* [1987] that color generally correlates with NRM intensity, so that gray to black corresponds to low-grade metamorphism, and the light color with higher CM.

A low mean destructive field (MDF), the alternating field necessary to erase half of the NRM (Table 1) is indicative of incomplete remagnetization [*Cisowski and Fuller*, 1987]. Our data

TABLE 1. Ore Minerals and Magnetic Properties of CM Rocks

Site	Sample	Rock Type	MDF mT	*Jo* A/m
A +	HAY1-5	chalk, Ni rich pyrite framboids, oxidized to Fe_2O_3 .	-	2e-04 to 2e-05
B +	HAY7-12	white chalk, Fe_2O_3 phase.	25	2.6e-01
C +	HAY13-18	white chalk, Fe_2O_3 phase.	15-20	6.1
D *	HAY19-24	calcite, larnite, $MgFe_2O_4$, apatite.	10-15	2.5
E *	HAY25-32	spurrite , brownmillerite, $MgFe_2O_4$, fine grains of a Fe phase.	10-35	2.1
F *	HAY33-39 Z1-11	calcite apatite, oxidized Fe-S phase bearing Cr, V, Ni, Zn, Ti.	25-35	2.1e-02 to 4.1e-03
G +	HAY40-44	black phosphorite, fine grains of a Fe phase (3μm), contain Mn, Ni, Zn, V.	35	6.4e-04
H #	HAY45-51	gehlenite, garnet, Ni or Ti bearing chromite.	15-100	20 to 2e-01
I +	HAY53-59 Z30-35,Z43-48	gray limestone concretion, fine grains of a Fe phase 3-5μm, abundant pyrite, some of it oxidized.	30-45	6.6e-03
J *	HAY80-85	pink chalk, fine grains of a Fe phase (Ti, V, S.-bearing).	-	7.3e-03
K *	K1-6	gray chalk, maghemite Fe_2O_3 .	15-10	4.5e-01
Z *	Z18-23,Z49-82	pink chalk, fine grains of a Fe phase (Ti, V, S.-bearing).	15 -30 or very high	5.4e-01 to 2e-04

+ Nonmetamorphic - according to field and mineralogical criteria.

* Low to medium grade

High grade metamorphic

MDF, mean destructive field; *Jo* , initial NRM intensity.

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TABLE 2. Paleomagnetic Field Directions of CM Rocks

Site	N	Dec.g	Inc.g	Dec.s	Inc.s	K	α_{95}
B	6	2.5	33.3	2.5	33.3	361	3.0
C	7	3.0	34.7	3.0	34.7	386	2.7
D	7	22.9	39.7	22.9	39.7	137	4.5
E	7	14.5	42.7	14.5	42.7	122	4.8
F - 1	10	0.4	47.6	24.8	43.0	889	1.8
F - 2	4	11.1	47.6	32.9	39.1	92	7.3
G	5	354.6	50.3	22.6	47.7	293	3.7
H	3	359.9	46.7	359.9	46.7	110	7.7
I - 1	6	9.6	51.9	44.7	42.9	83	6.3
I - 2	7	10.1	58.2	51.6	46.9	272	3.2
I - 3	7	356.4	53.6	39.0	50.1	442	2.5
J	4	35.5	67.1	60.4	58.7	604	2.8
K	6	171.3	-49.7	171.3	-49.7	21	12.6
Z - 1	8	4.5	47.0	20.9	47.6	162	3.9
Z - 2	6	356.1	56.7	19.9	58.6	347	3.1
Z - 3	9	28.1	63.1	51.8	56.6	37	8.7
Z - 4	12	41.7	42.2	51.8	34.2	359	2.1
Z - 5	5	360.0	39.5	12.8	41.8	37	10.3
Mean.g	18	8.1	49.1	-	-	40	5.2
Mean.s	18	-	-	25.1	46.9	26	6.6

N, number of samples; Dec.g and Inc.g, declination and inclination in geographic coordinates; Dec.s and Inc.s, declination and inclination in stratigraphic coordinates; K, precision parameter [*Fisher*, 1953]; and α_{95}, radius of 95% confidence about the mean

do not reveal any systematic behavior on the basis of the MDF. The new magnetic vector of sites G and I (limestone concretions and phosphorites which contain some sulfide, and therefore is not completely oxidized) was probably acquired either by CRM, i.e., slow oxidation of a sulfide to magnetite, [*Suk et al.*, 1990], or as pTRM, i.e, partial oxidation during the thermal event.

Magnetic Mineralogy

The main phases we were concerned with in this study were iron sulfides and oxides. Their characterization was achieved by a combination of XRD and electron microprobe. The AF and thermal demagnetization experiments together with IRM behavior were used to (1) evaluate whether the rock has been fully or partially remagnetized and (2) to establish the vector components and refine the definition of the magnetic carriers.

The electron probe analyses suggested the presence of a Fe_2O_3 phase as an important carrier. However, this phase could be either maghemite (γ-Fe_2O_3) or hematite (α-Fe_2O_3) or both. The two Fe_2O_3 phases are common constituents in different metamorphic rock types. Their most frequent occurrence is as maghemite-hematite nodules [*Gross*, 1977]. The orthogonal vector plot and the normalized intensity (J/J_0) plot of samples Z19 and Z23 in Figure 2 demonstrate very clearly the presence of both phases. The AF experiment (samples Z19A and Z23A) shows two distinct coercivity spectra: a low to medium coercivity component and a very high coercivity component. The thermal demagnetization experiment (Figure 2, samples Z19B and Z23B) shows two sharp decreases in intensity, one at about 300°C, the second at 670°C. The first is typical for maghemite, which is stable at atmospheric pressures up to 300°C - 400°C, and the second for hematite with a Curie temperature of about 670°C. The single component vector and the presence of the two mineral phases suggest that both phases were formed in a single event. Apparently, combustion attained temperatures of 300°C - 400°C. At this temperature, maghemite was dominant but probably started to convert to hematite. Samples Z-59, Z-60, and Z-67 (Figure 3) show only one phase with a Curie temperature of 670°C, which is α-Fe_2O_3 . This suggests that rocks at this site have been heated to temperatures higher than 400°C. It is difficult to place an upper limit on the temperature of metamorphism, since calcite is the dominant phase in these rocks. In a pure limestone calcite is stable up to about 800°C, when it undergoes decarbonization. The situation is further complicated by the appearance of recarbonated calcites, i.e. carbonates which underwent decarbonation in the prograde stage, but reabsorbed CO_2 during cooling and weathering. Fortunately these calcites can be distinguished from the relict ones both texturally and by their isotopic composition [*Kolodny and Gross*, 1974]. The main magnetic carrier of samples HAY8 and HAY15 (Figure 4) is maghemite, although a minor amount of another phase (probably magnetite) is present. Samples HAY24 (Figure 5), HAY26, and HAY27 (Figure 6) contain magnesioferrite with an intensity drop at about 400°C; the reported Curie temperature for magnesioferrite is 440°C [*Piper*, 1987, p. 30].

The nature of the iron phases in different metamorphic grades is of major importance in interpreting the rock-magnetic and paleomagnetic results. The iron-bearing phase in the nonmetamorphic protoliths is mainly a sulfide, probably pyrite. The oxidation of sulfide is among the earliest manifestations of metamorphism. Many of the medium grade rocks are spotted by maghemite or maghemite-hematite concretions, pseudomorphic

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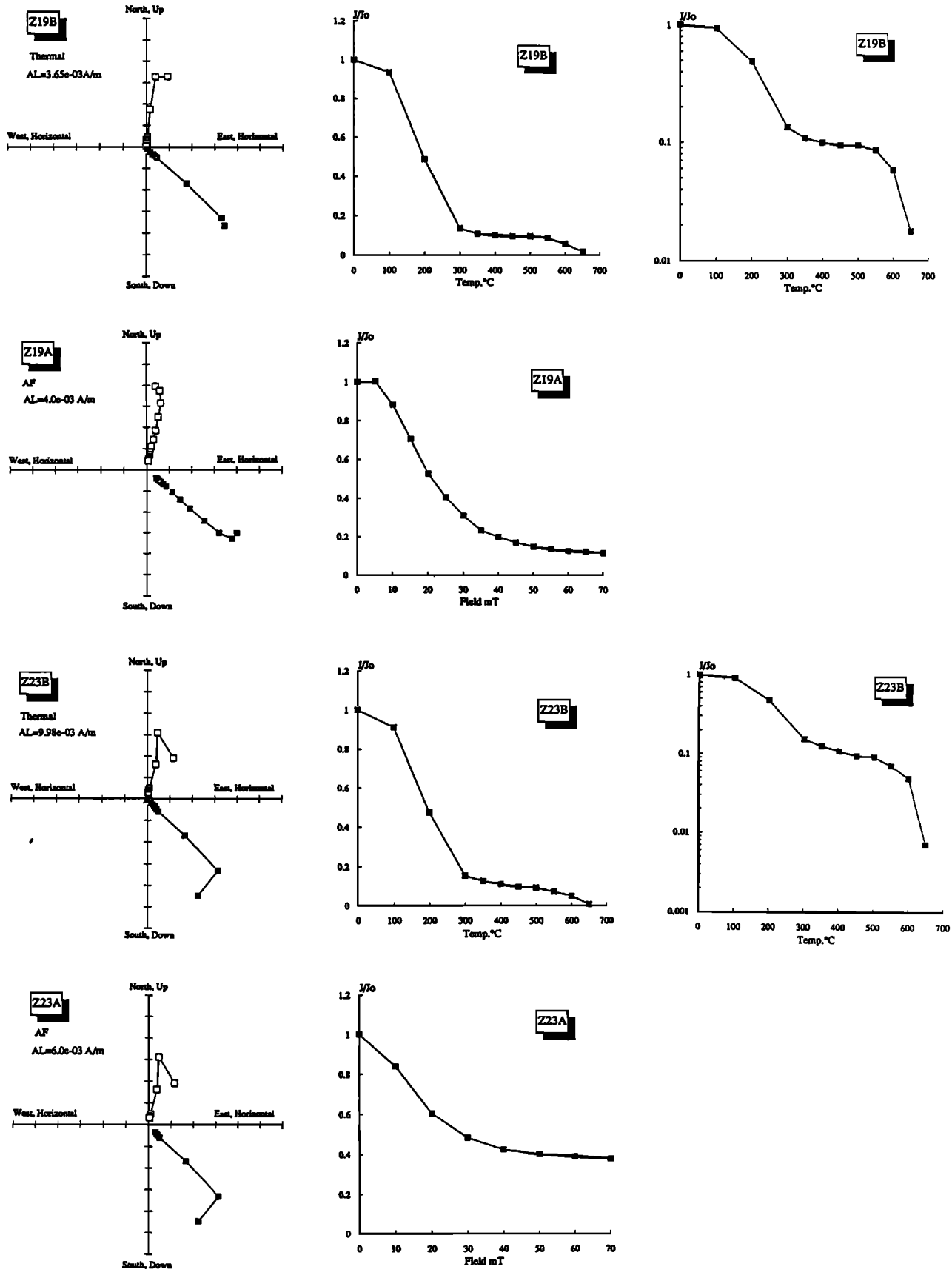


Fig. 2. Orthogonal vector plot of both AF and thermal demagnetization experiments of two samples demonstrates the vector stability. The normalized intensity plots J/J_0 (thermal demagnetization experiment presented both on linear and semi-log plots) show the transition temperature (about 300°C) of maghemite and the curie temperature of hematite (about 670°C). The plot demonstrates that both phases are present.

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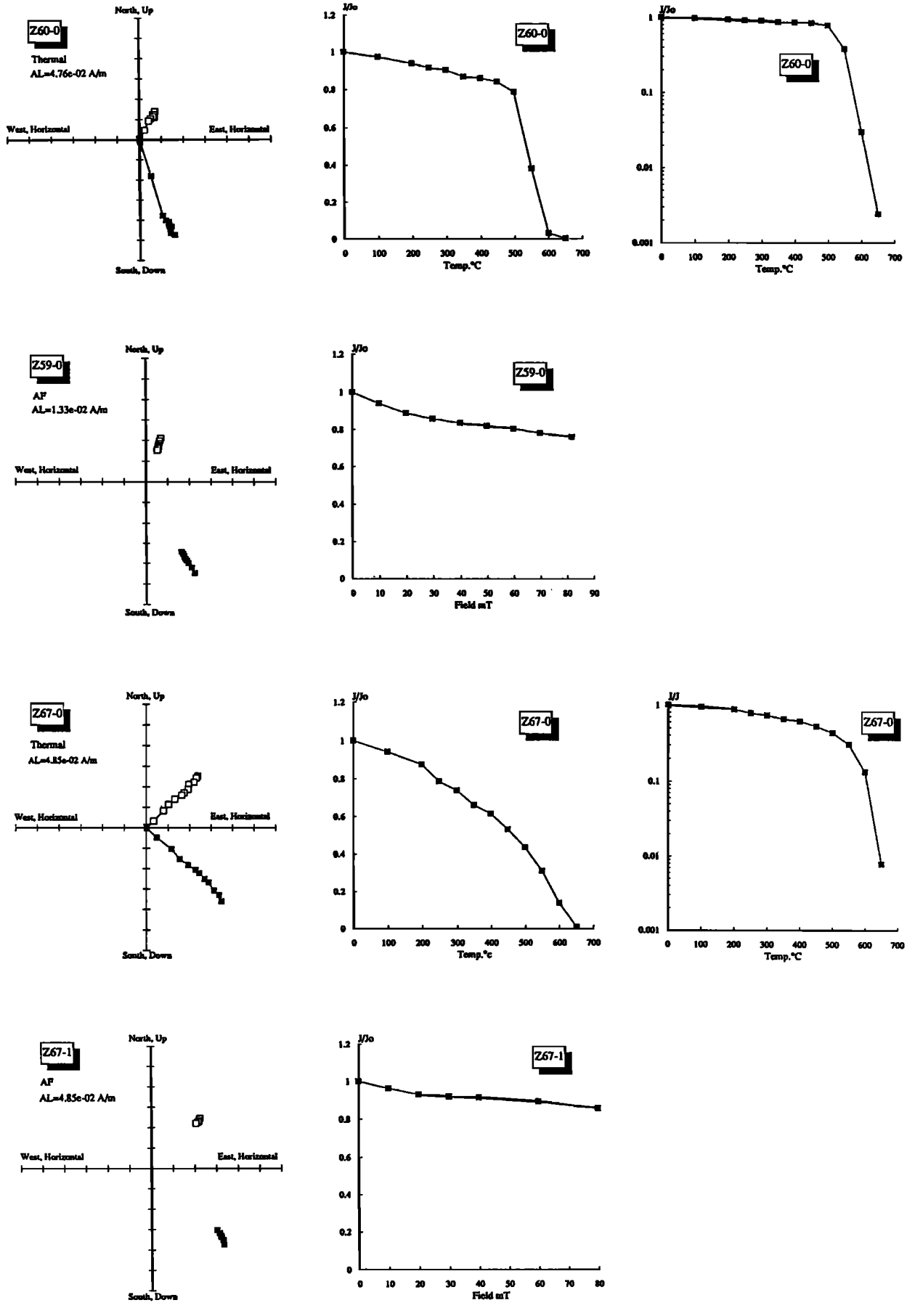


Fig. 3. Orthogonal vector plot and normalized intensity plot J/J_0 (thermal demagnetization experiment presented both on linear and semi-log plots) of CM samples which contain mostly hematite. The plot implies burning temperatures higher than 400°C.

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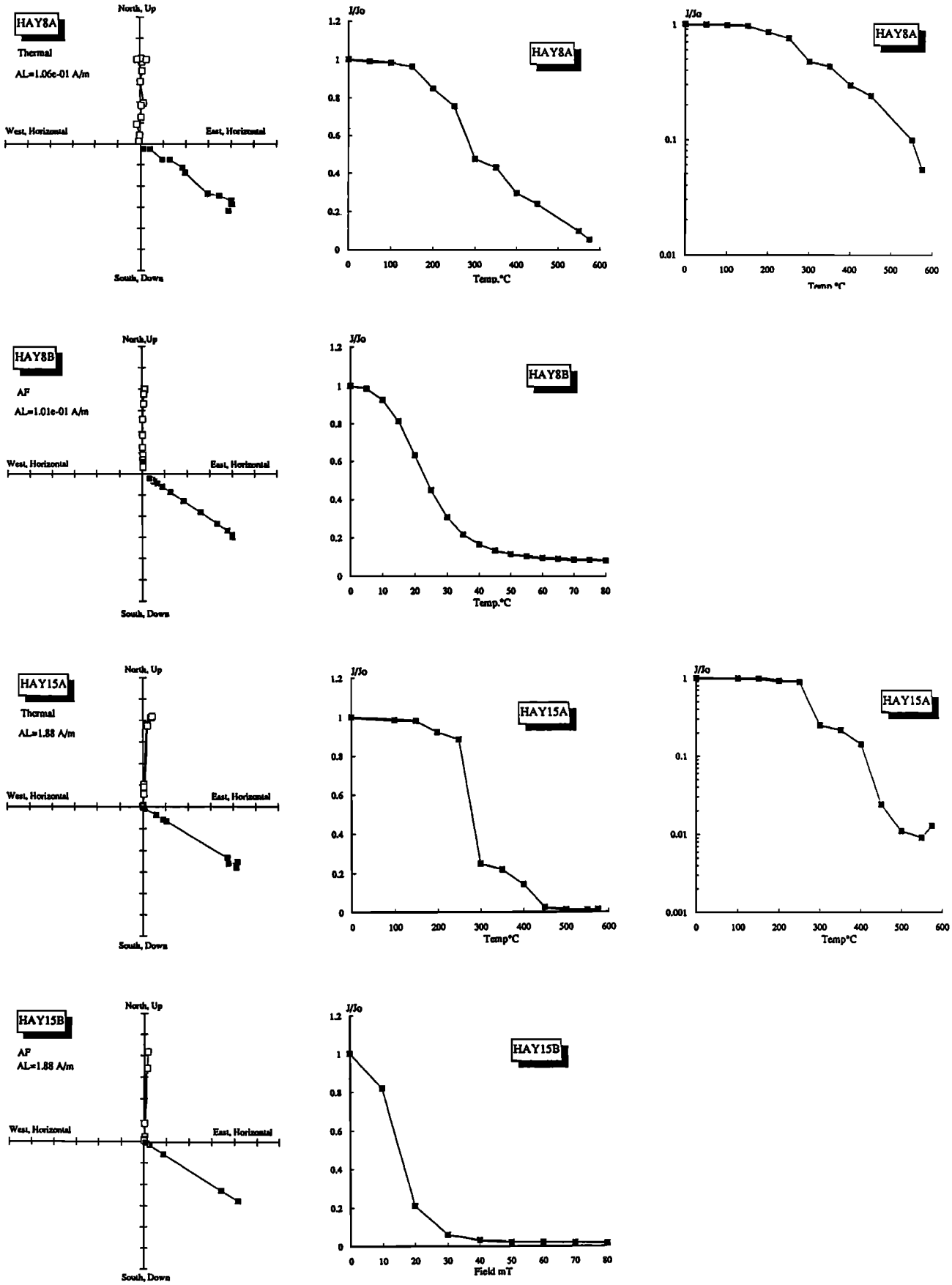


Fig. 4. Orthogonal and normalized intensity plots J/J_0 (thermal demagnetization experiment presented both on linear and semi-log plots) of both AF and thermal demagnetization experiments of samples HAY 8 and HAY 15. Both specimens were sampled as "unheated" rocks, based on field, mineralogical and stable isotopic criteria. The plots demonstrate the complete remagnetization and the presence of maghemite as the magnetic carrier.

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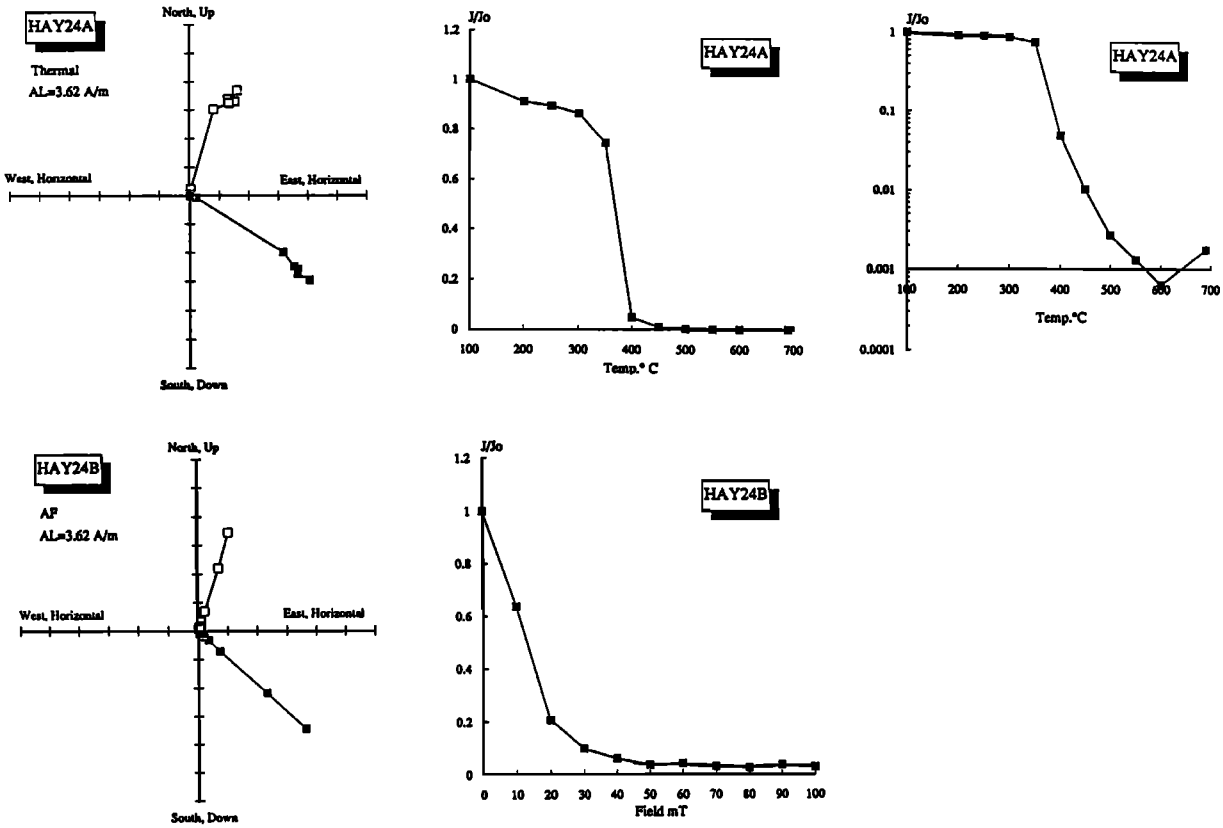


Fig. 5. Orthogonal vector and normalized intensity plots J/J_0 (thermal demagnetization experiment presented both on linear and semi-log plots) of medium grade CM rocks which contain magnesioferrite ($MgFe_2O_4$) as remanence carrier.

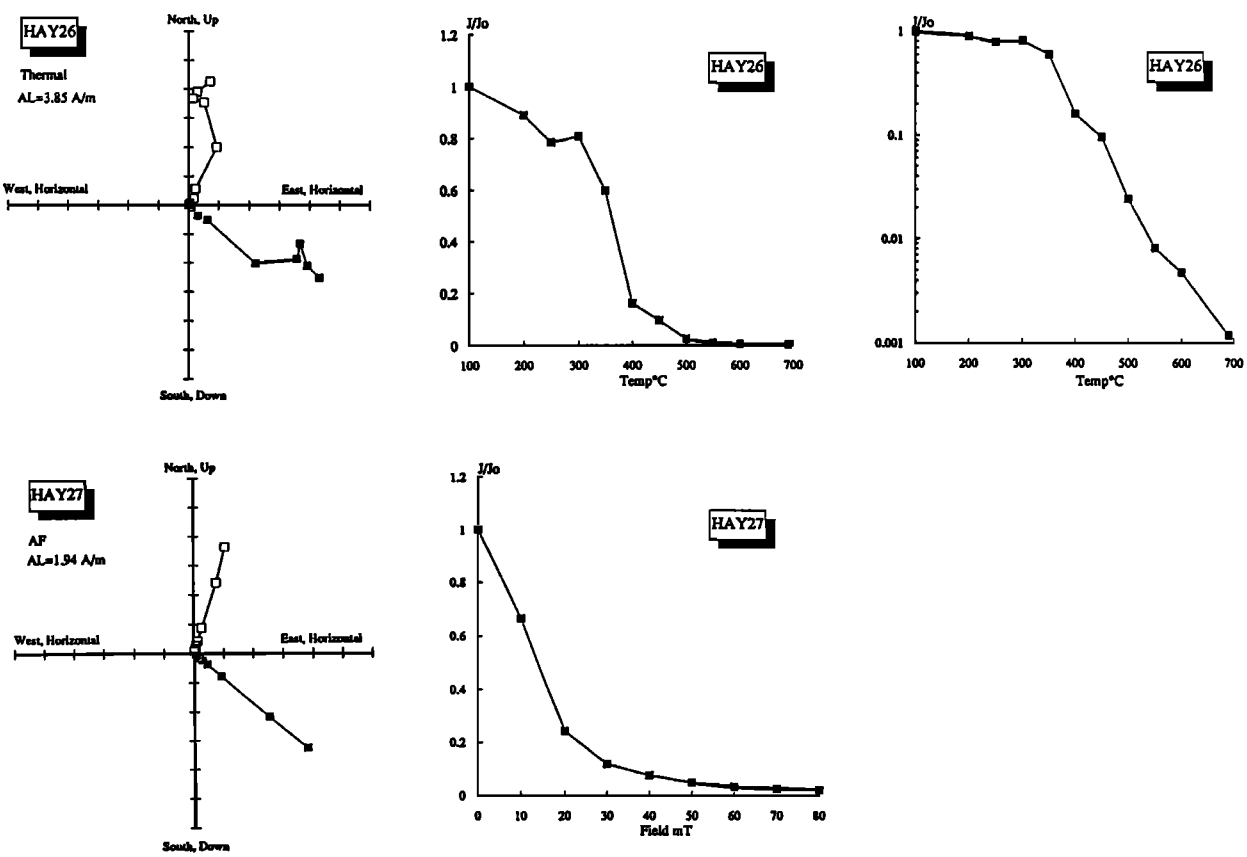


Fig. 6. Orthogonal vector and normalized intensity plots JJ/J_0 (thermal demagnetization experiment presented both on linear and semi-log plots) of medium-grade CM rocks which contain magnesioferrite ($MgFe_2O_4$) as remanence carrier.

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after the pyrite concretions of the protolith (Figure 7). We did not succeed in resolving the spatial and textural relations between the two Fe_2O_3 polymorphs. EPM analysis shows that framboidal textures which are unmistakably indicative of pyrite, are in fact made up of an iron oxide, most likely hematite (Figure 8). A similar transformation of framboidal pyrite into iron oxides has been observed and discussed in the paleomagnetic context by *Suk et al.* [1990].

As evident from the previous description, maghemite (γ-Fe_2O_3) is a very abundant phase in MZ rocks. *Gross* [1977] considered maghemite to be a retrograde mineral, suggesting that it is "probably an alteration product of magnetite." Such alteration seems, however, to be efficient only at temperatures above about 100°C [*Basantsev*, 1976]. A study by *Cole et al.* [1987] on the behavior of iron minerals during coal heating in air may provide direct insight as to the origin of maghemite. The pyrite originally present in coal is first oxidized to ferrous sulphate. At about 325°C, maghemite is formed, which is transformed into hematite (α-Fe_2O_3) between 350°C and 400°C. The heating of coal in air seems like a very good simulation of the processes taking place during the metamorphism of the bituminous chalks and marls of the MZ. The prograde origin of maghemite is supported not only by the coal-heating experiment, but also by the detection of relict sulfide in some of the low-grade rocks and of iron oxide pseudomorphs after framboidal sulfide and by the paragenesis of maghemite and hematite. XRD analysis suggests that most of the relict sulfide in the moderately metamorphic rocks is greigite (Fe_3S_4), which is a ferrimagnetic phase. No evidence for greigite has been obtained from our magnetic studies. The presence of maghemite may indicate some upper temperature limit for the rock in question, though the precise value of this limit is not clear. The γ-Fe_2O_3 to α-Fe_2O_3transition seems to be dependent on the previous history of the maghemite and its grain-size. Some maghemites are stable with respect to hematite at temperatures as high as 500°C, complete inversion of synthetic maghemite being achieved only at 660°C [*Ozdemir and Banerjee*, 1984].

On the other hand, the paragenesis of maghemite with high-temperature minerals such as gehlenite (and magnetite) may suggest that in those cases, maghemite is indeed a retrograde phase, the result of magnetite alteration. The nature of maghemite may thus not be unique: the maghemite-hematite paragenesis may be prograde, whereas maghemite-magnetite, retrograde. The identity of the direction of magnetic vectors of different magnetic phases (as discussed above) suggests that, for the purpose of our discussion, the solution of the "prograde-retrograde" dilemma may not be crucial: apparently, the entire metamorphic process which resulted in the formation of the different iron oxides proceeded in the same magnetic field. Even though metamorphism at different sites may have occurred at different times, in each locality the entire process, from ignition to the peak of metamorphism, to the cooling and retrograde reactions was rather short.

The IRM experiment of samples HAY20 and HAY29 (Figure 9) which contain maghemite show saturation in a field of 75 to 200 mT, which is also indicative for this mineral [*Ozdemir and Banerjee*, 1984]. Samples HAY9 and HAY17 have not fully saturated (Figure 9). This behavior is compatible with the thermal demagnetization and EPM results, which indicate both maghemite and hematite as magnetic carriers. Phosphorites and limestone concretions (samples Z48 and HAY40, sites I and G), which are only a few tens of meters away from in situ CM rocks, show no mineralogical or petrographic signs of metamorphism. Their magnetic vectors, however, strongly suggest almost complete remagnetization during the Late Neogene, as well as coercivity spectra and Curie temperature typical for magnetite (Figure 10). We suggest that these phosphorites and micritic limestones were heated to low temperatures with minor amounts of oxygen, which led to growth of magnetite and acquisition of partial thermal remanence (pTRM). The presence of pyrite is indicative of this incomplete process. However, we cannot ignore the possibility that this vector is a chemical remanent magnetization (CRM) acquired during low-temperature remagnetization [*Suk et al.* 1990]. The IRM experiment of this sample is also compatible with

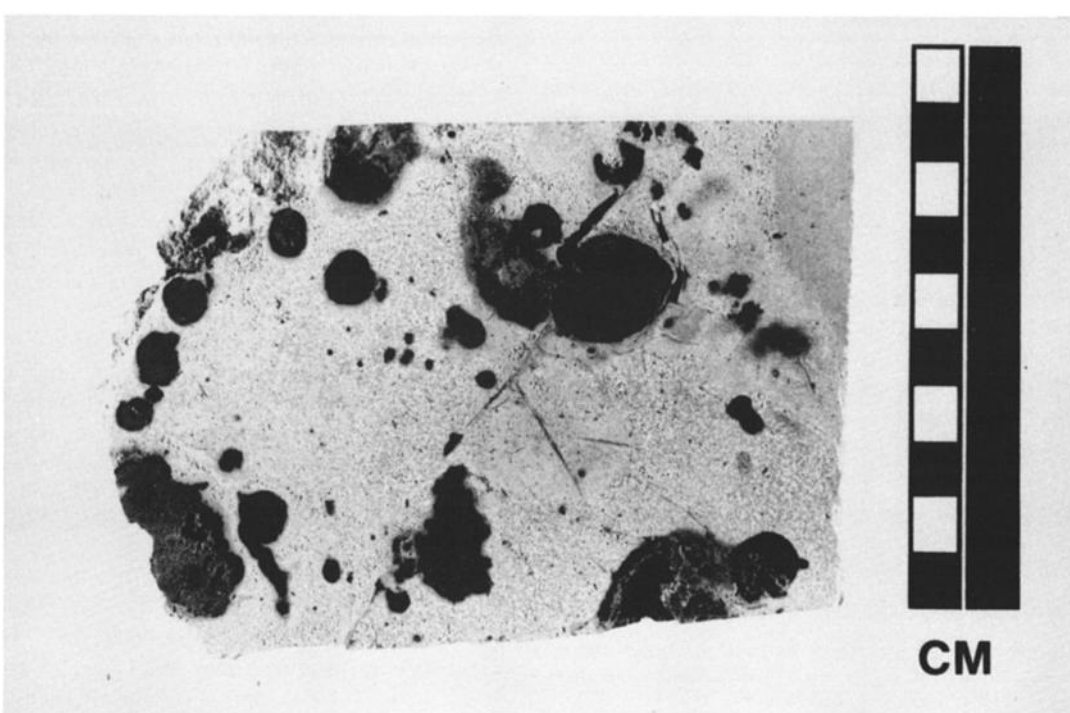
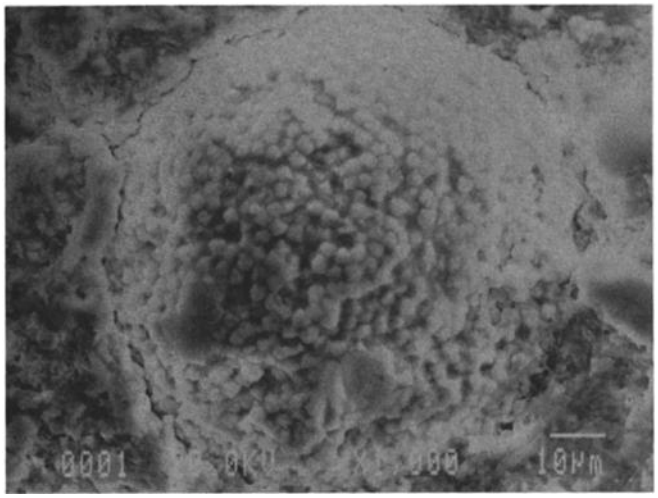
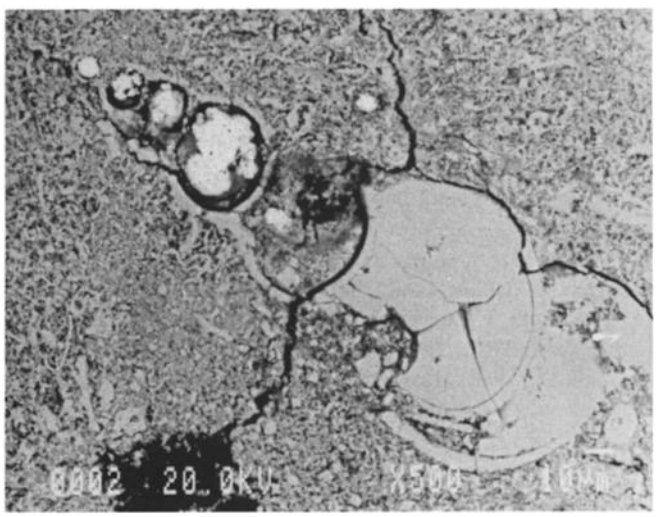


Fig. 7. Maghemite-hematite nodules in diopside-zeolite CM rock.

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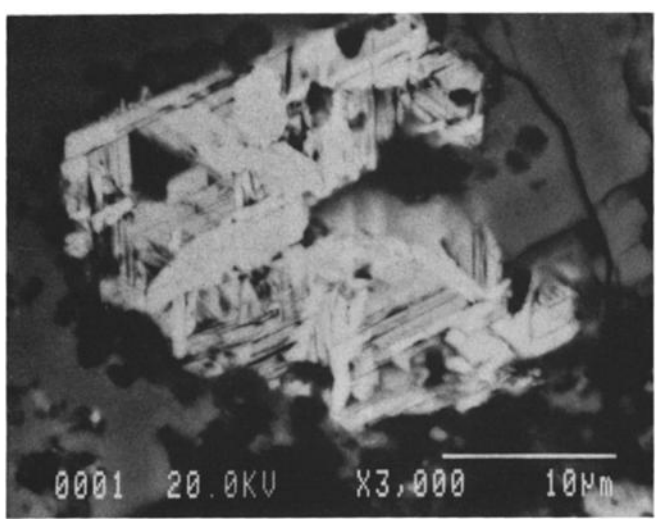


Fig. 8. (*a*) Fe_2O_3 pseudomorph after sulfide framboid (secondary electrons image). (*b*) Maghemite infilling in fossil foraminifer (back scattered electron image). (*c*) Magnesioferite (back-scattered electron image).

magnetite being the principal magnetic carrier, though a minute amount of hematite is present.

Directions and Time of Remagnetization

The Late Cretaceous and the Late Neogene poles position of the stable African plate were calculated [*Piper* 1988] to predict direction of magnetization for the study area. It is clear from Table 3 that the two expected directions of magnetization (D=348.5°, I = 17° for Late Cretaceous and D = 6.1°, I = 50° for the Pliocene) are very different, and therefore it is relatively simple to distinguish between them. Most of the samples show stable behavior during demagnetization and usually a single component vector of magnetization (Figures 2-6 and 10), with a very low H_C overprint of the present field in some cases (Figure 2 and 10). The mean directions for all site means are summarized in Table 3, both in geographic and stratigraphic coordinates. The steep inclination and northerly oriented declination even for sites which are considered to be nonmetamorphic (sites B, C, G, and I; see Tables 1 and 2 for comparison, samples HAY8 and HAY15 in Figure 4, and Z48 and HAY40A in Figure 10) suggest that the rocks from all our sampling sites have been completely remagnetized and acquired new thermoremanent magnetization (TRM) or CRM as a result of the CM process during the Late Neogene. The declination and inclination of the geographic mean (8.2° / 49.2°) versus the declination and inclination of the stratigraphic mean (25.1° / 46.9°), together with higher K and smaller α_{95} of the geographic mean, suggest that all CM rocks acquired their magnetization after folding and tilting. Sites B, C, D, E, and H were sampled along a traverse from a nonmetamorphic zone through a low and medium grade to a high CM zone. These sites show different inclination and declination with small α_{95}. If these changes are not the result of some unrecognized structural features (landslides, collapses, etc.), then they may reflect paleosecular variation of the Earth's magnetic field during the cooling of the burning rock pile. The dual polarities are clear evidence that ignition and the CM process took place during different times at different localities. This result is compatible with the different ages obtained by geochronological methods for the CM event at different sites. If all obtained ages are indeed correct, a time difference of more than 10 million years elapsed between ignition at Maale Adumim and the Hatrurim basin. A similar behavior pattern has been noted by *Jones et al.*, [1984], who measured both normal and reverse polarities in a Pleistocene clinker deposit in the Powder River Basin in Wyoming and Montana.

All sampled sites at Hatrurim and Maale Adumim are located on the flanks of the Judea anticlines and not far from the Dead Sea transform. Although this part of the plate margin is very close to the transform the northerly geographic (in situ) declination and inclination (D=8.2°, I=49.2°) suggest that CM outcrops and the adjacent area have not been rotated about either a horizontal or a vertical axis since the time of the burning event (Table 3, Figure 11). If indeed the burning at Maale Adumim occurred in the Late Miocene, then this result implies that this area has not been significantly deformed since that time, either by the Syrian arc folding or by the faulting and rifting associated with the Dead Sea Transform.

SUMMARY

As is evident from the geological map, the CM process affected bituminous rocks in numerous small outcrops as well as several larger areas. The burning has completely changed the magnetic

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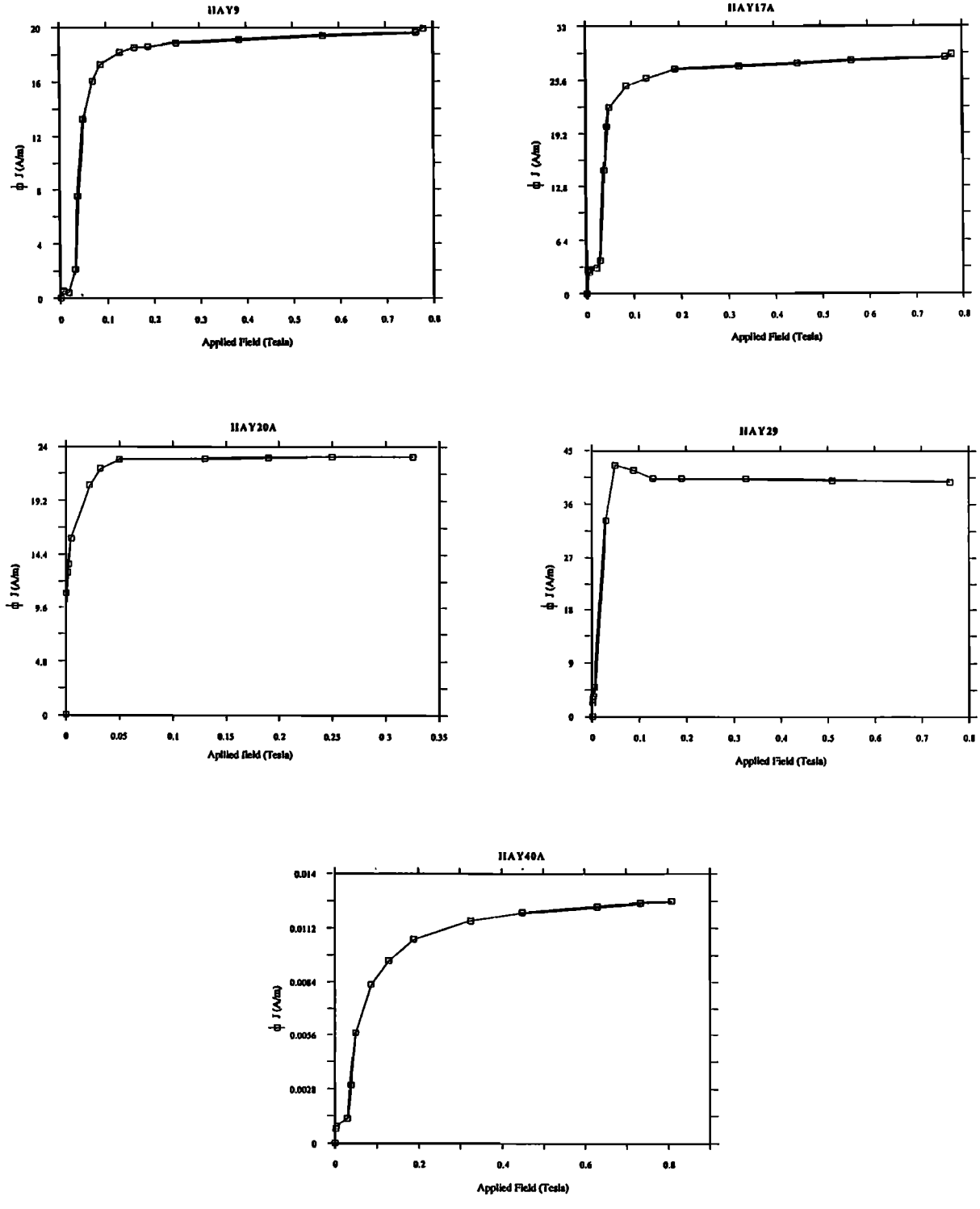


Fig. 9. IRM experiments of some selected CM rocks (see Table 1).

mineralogy and properties of the bituminous protolith. The most important Fe phase in the protolith is pyrite, which during oxidation to high susceptible magnetic material appears to be responsible for much of the observed magnetic enhancement. Sulfide oxidation was also a possible source of heat for the initiation of spontaneous combustion.

Mineralogical analysis indicates that abundant new magnetic materials were produced during burning. The AF and thermal demagnetization experiments indicate that the most important new oxide is Fe_2O_3 both as maghemite (γ-Fe_2O_3) and hematite (α-Fe_2O_3), with two implications. First, the burning took place in a well-aerated environment. Second, the transition temperature of maghemite to hematite (300°-400°C) may serve as a maximum temperature indicator.

The declination and inclination of the new TRM are compatible with the late Neogene geomagnetic field for this area. This

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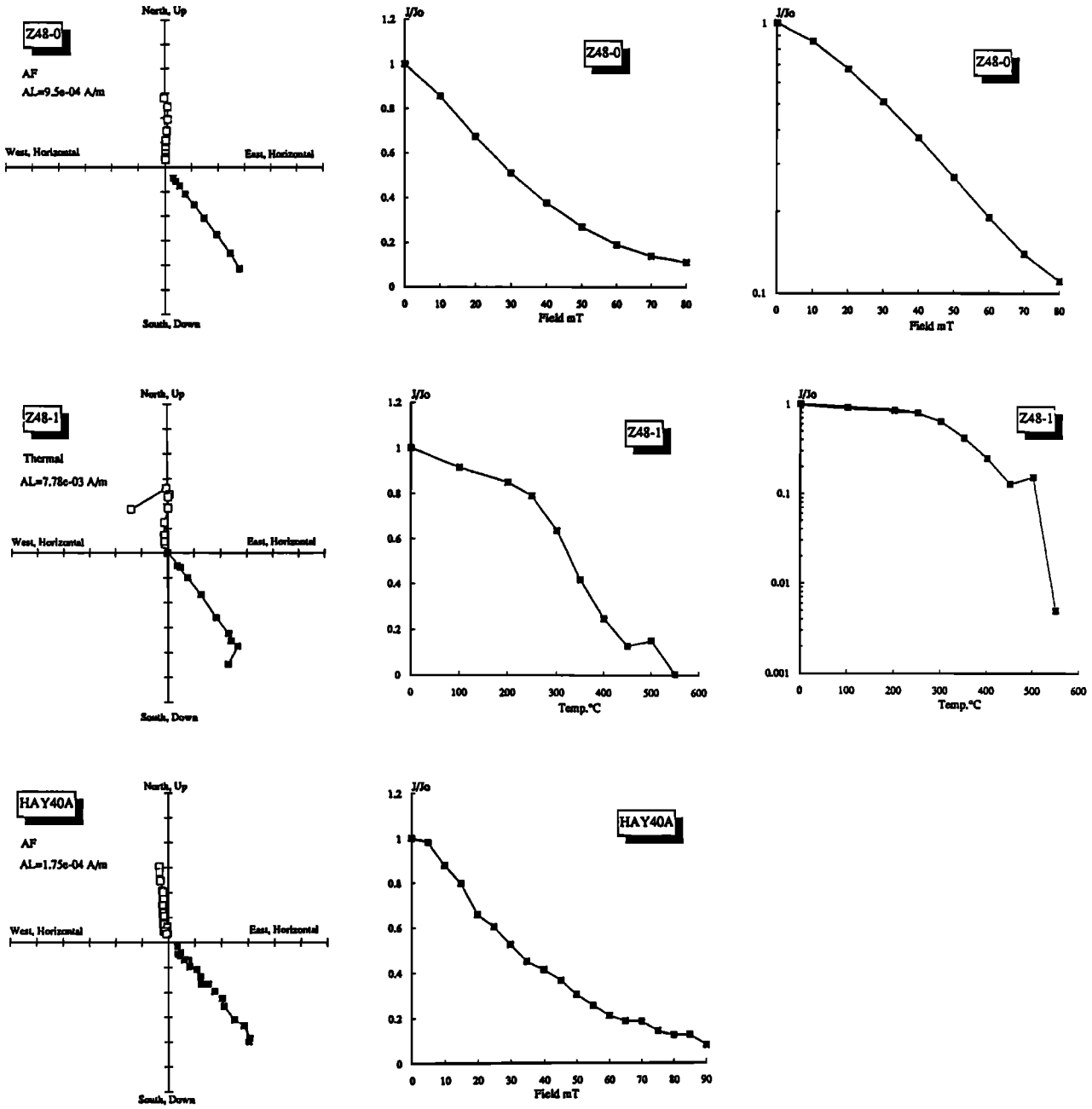


Fig. 10. Orthogonal vector and normalized intensity plots J/J_0 (thermal demagnetization experiment presented both on linear and semi-log plots) of phosphorite and limestone concretions. The rocks were sampled as "unheated" about 200 m away from the CM outcrop. Remagnetization, by pTRM or CRM, with magnetite as the new remanence carrier is demonstrated.

TABLE 3. Mean Paleomagnetic Vectors and Poles from CM Rocks and Mean Poles Derived from Stable Africa

Age	*N*	Dec.g	Inc.g	Dec.s	Inc.s	Kg/Ks	α_{95} (g/s)	Long.°E	Lat.°N
Upper Cretaceous	9	348.5	17.0	-	-	-	5.4	243.0	64.8
Miocene	21	9.2	47.3	-	-	-	3.3	144.0	81.1
Pliocene	17	6.1	50.0	-	-	-	1.7	132.2	84.7
Arad area	11	6.1	46.3	22.8	43.6	69/34	5.1/7.2	160.1	83.5
Adumim area	6	16.2	54.0	35.6	51.1	24/24	11.6/11.6	120.0	78.3
All CM sites	18	8.2	49.2	25.1	46.9	40/26	5.2/6.5	135.6	82.9

N, number of poles or sites; Dec.g and Inc.g, declination and inclination in geographic coordinates; Dec.s and Inc.s, declination and inclination in stratigraphic coordinates; Kg/Ks, geographic and stratigraphic precision parameter [*Fisher*, 1953]; α_{95}, radius of 95% confidence about the mean; Long. = 35.3° E and Lat. = 31.5° N.

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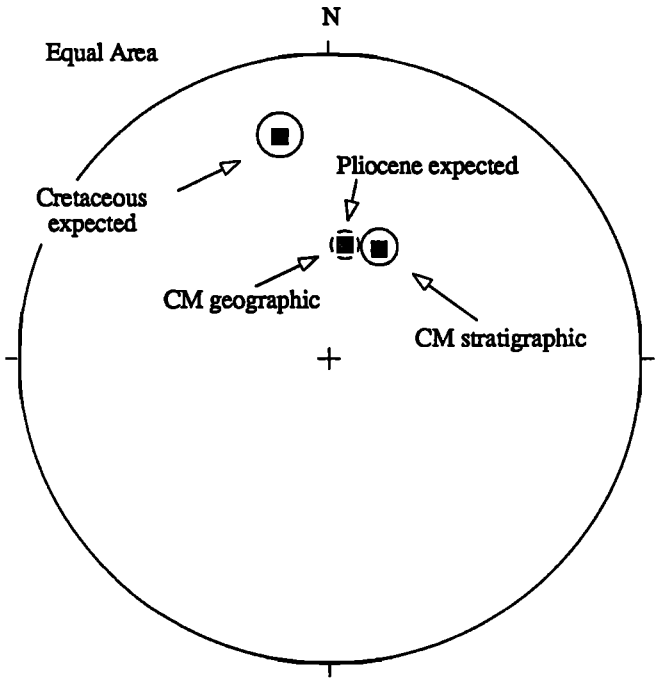


Fig. 11. Lower hemisphere stereographic projection of all sites mean geographic and stratigraphic directions of paleomagnetic vectors, and the expected field directions for the Late Cretaceous and Pliocene times. The stratigraphic mean of CM rocks best correlates with the Pliocene expected field direction. Small ovals are the 95% confidence area about the mean.

conclusion is consistent with all obtained radiometric ages of CM rocks. The northerly geographic declination and the steep geographic inclination imply that this area has not been significantly deformed since the time of burning.

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REFERENCES

Basantsev, G. P., Temperature conditions for maghemite (Gamma hematite) formation, *Izv. Akad. Nauk SSSR (Russian Metallurgy)*, no.4, 33-35, 1976.

Bentor, Y. K., S. Gross, and L.Heller, High temperature minerals in non-metamorphosed sediments in Israel, *Nature, 199,* 478-479, 1963.

Bentor, Y. K., M. Kastner, I. Perlman, and Y. Yellin, Combustion metamorphism of bituminous sediments and the formation of melts of granitic and sedimentary composition, *Geochim Cosmochim. Acta, 45,* 2229-2255, 1981.

Bentor, Y. K., and A. Vroman, The Geological map of Israel, scale 1:100,000, sheet 16, Mount Sedom (with explanatory text), Geol. Surv., of Israel, Jerusalem, 1960.

Burg, A., A. Starinsky, Y. Kolodny, and Y. Bartov, The geology of the "mottled zone" (Hatrurim Formation) in the Hatrurim Basin, *Isr. J. Earth Sci.*, in press, 1992.

Cisowski, M S., Magnetic properties of K/T and E/O microspherules: origin by combustion?, *Earth Planet. Sci. Lett., 88,* 193-208, 1988.

Cisowski, M.S., and M. Fuller, The generation of magnetic anomalies by combustion metamorphism of sedimentary rock, and its significance to hydrocarbon exploration, *Geol. Soc. America Bull., 99,* 21-99, 1987.

Cole, D. A., G. W. Simmons, R. G. Herman, K. Klier, and I. Czako-Nagy, Transformation of iron minerals during coal oxidation, *Fuel, 66,* 1240-1248, 1987.

Fisher, R. A., Dispersion on a sphere, *Proc. R. Soc. London, Ser. A, 217,* 295-305, 1953.

Folkman, Y., and Z. Yuval (compilers), Israel Aeromagnetic map; scale 1:250.000, Institute of Petrolum Research and Geophysics, Israel Geological Survey, Jerusalem, 1976.

Gross, S., The mineralogy of the Hatrurim Formation, Israel, *Isr. Geol. Surv. Bull., 70,* 80 pp., 25 pl., 1977.

Hasbrouck, W. P. and F. A. Hadsell, Geophysical exploration techniques applied to western United States coal deposits, in *First International Coal Exploration Symposium, Proceeding*, Miller Freeman, San Francisco, Pp. 256-287, Calif., 1976.

Jones, A. H., J. W. Geissman, and D. A. Coates, Clinker deposits, Powder River basin, Wyoming and Montana: a new source of high-fidelity paleomagnetic data for the Quaternary, *Geophys. Res. Lett., 11,* 1231-1234, 1984.

Kirschvink, J. L., The least squares line and plane and the analysis of paleomagnetic data, *Geophys. J. R. Astron. Soc., 62,* 699-718, 1980.

Kolodny, Y., M. Bar, and E. Sass, Fission track age of the "Mottled Zone event", Israel, *Earth Planet. Sci. Lett., 11,* 269-272, 1971.

Kolodny, Y., and S. Gross, Thermal metamorphism by combustion of organic matter, isotopic and petrological evidence, *J. Geol., 82*, 489-506, 1974.

Kolodny, Y., N. Schulman, and S. Gross, Hazeva Formation sediments affected by "Mottled Zone event", *Jsr. J. Earth Sci., 22,* 185-193, 1973.

Lees, G. M., The chert beds of Palestine, *Proc. Geol. Assoc. London, 39,* 445-462, 1928.

Mathews, W. H., and R. M. Bustin, Why do the Smoking Hills smoke?, *Can. J. Earth Sci., 21,* 737-742, 1984.

Matthews, A., and S. Gross, Petrologic evolution of the Mottled Zone (Hatrurim) metamorphic complex of Israel, *Isr. J. Earth Sci., 29,* 93-106, 1980.

Matthews, A., and Y. Kolodny, Oxygen isotope fractionation in decarbonation metamorphism: "The Mottled Zone event", *Earth Planet. Sci. Lett., 39,* 179-192, 1978.

Matthews, A., and Y. Nathan, The decarbonation of carbonate-fluorapatite (Francolite), *Am. Mineral., 62,* 565-573, 1977.

Ozdemir, O., and S. K. Banerjee, High temperature stability of maghemite (γ-Fe_2O_3), *Geophys. Res. Lett., 11,* 161-164, 1984.

Piper, J. D. A., *Paleomagnetism and the Continental Crust*, 434 pp., Open University Press and Haksted Press, John Wiley New York, 1987.

Piper, J. D. A., *Paleomagnetic Database*, 264 pp., Open University Press, Milton Keynes, 1988.

Reverdatto, V. V., *The Facies of Contact Metamorphism* 263 pp., Australian National University, Canberra, 1970. (Translated from Russian, 1973.)

Reynolds, R. L., A polished view of remagnetization, *Nature, 345*, 579-580, 1990.

Suk, D., D. R. Peacor and R. Van der Voo, Replacement of pyrite framboids by magnetite in limestone and implications for paleomagnetism, *Nature, 345,* 611-613, 1990.

Watson, D. E., Magnetic properties of clinker, *Geophysics, 44*, 376- 377, 1978.

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